

Software Requirements Specification

On

ONLINE FOOD ORDERING **SYSTEM**

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1. INTRODUCTION

1.1 Purpose

The purpose of this SRS is to define the requirements, users, stakeholders, and expected behavior of the **Online Food Ordering System**. It provides a common reference for development and testing of the system.

1.2 Scope

The system allows customers to browse restaurants and menus, place food orders, make payments, and track deliveries. Restaurants can manage menus and orders, while delivery personnel can manage assigned deliveries.

The system does not directly control physical food preparation or delivery.

1.3 Definitions

Term	Definition
Customer	User who orders food through the system
Restaurant	Food provider registered on the platform
Delivery Personnel	Person responsible for delivering orders
Admin	Person responsible for managing the system
Order ID	Unique identifier assigned to an order

2. STAKEHOLDERS AND USER ROLES

Stakeholders are individuals, organizations, or external systems that interact with or are affected by the Online Food Ordering System.

Stakeholder	Role / Interest	Major System Requirements
Customer	Orders food and tracks deliveries	Search, cart, ordering, payment, tracking, reviews
Restaurant Owner/Staff	Manages restaurant and prepares orders	Menu management, order acceptance, availability, sales
Delivery Personnel	Delivers customer orders	Assigned orders, delivery details, status updates
System Administrator	Maintains and controls the platform	User management, restaurant management, monitoring, security
Management	Monitors business performance	Revenue reports, order analytics, performance statistics

Payment Gateway	Processes online transactions	Payment processing and transaction verification
Notification Service	Sends system notifications	Order, payment, and delivery notifications

Stakeholder Priorities

- **Customers** require convenience, accurate information, secure payment, and timely order updates.
- **Restaurants** require efficient order and menu management.
- **Delivery personnel** require accurate delivery and order information.
- **Administrators** require control, security, and monitoring capabilities.
- **Management** requires reports and analytics for decision-making.
- **External services** require secure and reliable communication with the system.

3. FUNCTIONAL REQUIREMENTS

3.1 User Authentication

- The system shall allow users to register and log in securely.
- The system shall provide role-based access.
- Users shall be able to update their profile information.

3.2 Restaurant & Menu Management

- Restaurant staff shall be able to add, update, and remove food items.
- Staff shall be able to update prices and food availability.
- Customers shall be able to view restaurant menus.

3.3 Food Search & Ordering

- Customers shall be able to search restaurants and food items.
- Customers shall be able to filter restaurants by cuisine, price, and rating.
- Customers shall be able to add items to a cart and modify quantities.
- The system shall calculate the total order amount.
- The system shall generate a unique Order ID.

3.4 Payment & Order Tracking

- The system shall support online payment methods.
- The system shall record payment status.
- Customers shall be able to view their order status.

- Restaurants shall be able to accept or reject orders.
- Delivery personnel shall be able to update delivery status.

3.5 Reviews & Notifications

- Customers shall be able to rate and review completed orders.
 - The system shall notify users about important order-status changes.
 - Administrators shall be able to moderate inappropriate reviews.
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4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance

- Normal requests should be processed within 2 seconds.
- The system should support at least 500 concurrent users.

4.2 Security

- Passwords shall be stored using secure hashing algorithms.
- Role-based authorization shall prevent unauthorized access.
- Payment communication shall use secure protocols.

4.3 Reliability

- The system shall maintain accurate order and payment records.
- Regular database backups shall be performed.
- System failures shall not cause duplication or loss of confirmed orders.

4.4 Usability

- The interface shall be simple and intuitive.
 - The system shall be responsive across desktops, tablets, and smartphones.
 - Error messages shall be clear and understandable.
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5. FEASIBILITY STUDY

5.1 Technical Feasibility

The system can be developed using commonly available technologies such as HTML, CSS, JavaScript/React, Node.js/Django, and MySQL/PostgreSQL/MongoDB. Payment and notification APIs can also be integrated.

5.2 Economic Feasibility

Open-source technologies can reduce development and licensing costs. The system can reduce manual work and improve restaurant operational efficiency.

5.3 Operational Feasibility

The system supports the existing workflow of customers, restaurants, and delivery personnel through dedicated interfaces. Minimal training would be required for regular users.